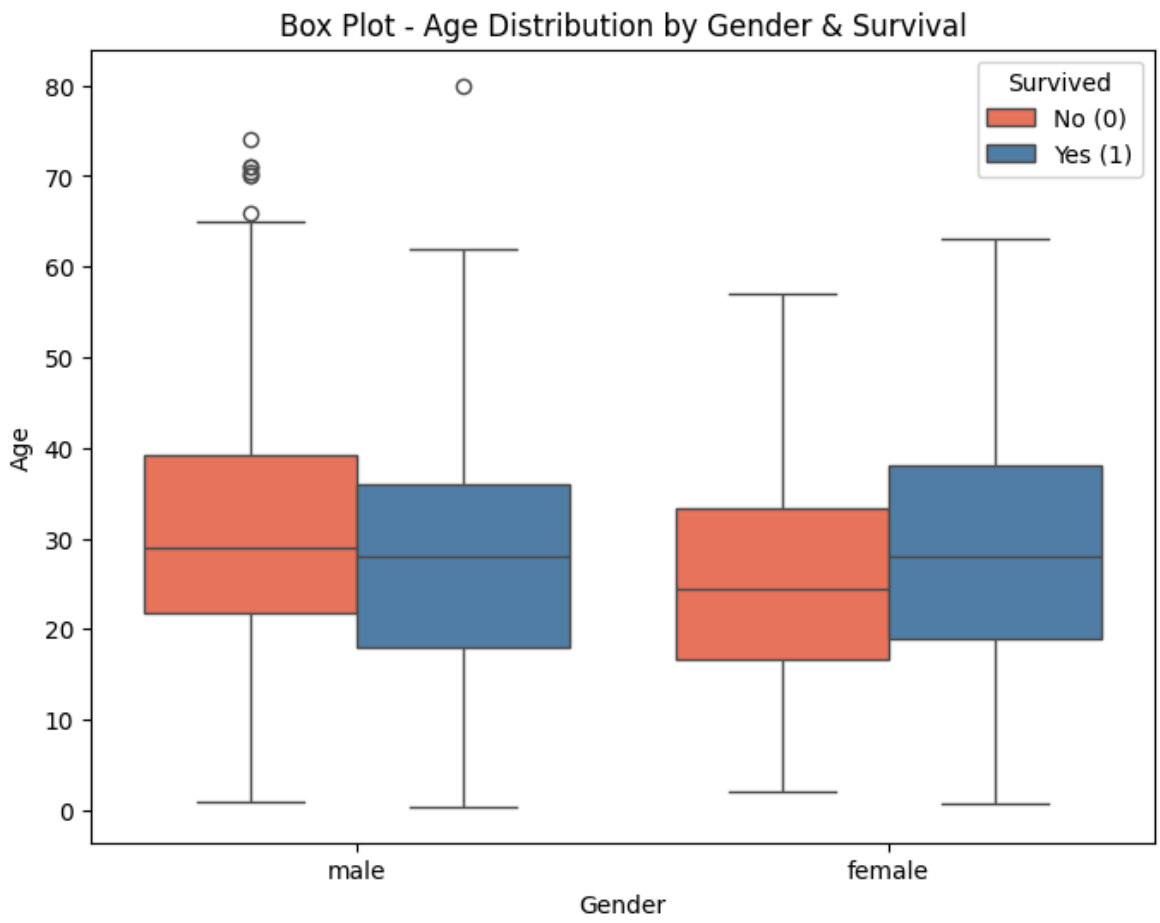
In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
dataset = sns.load_dataset('titanic')
print(dataset.head())
print("\nShape:", dataset.shape)
print("\nColumns:", dataset.columns.tolist())
```

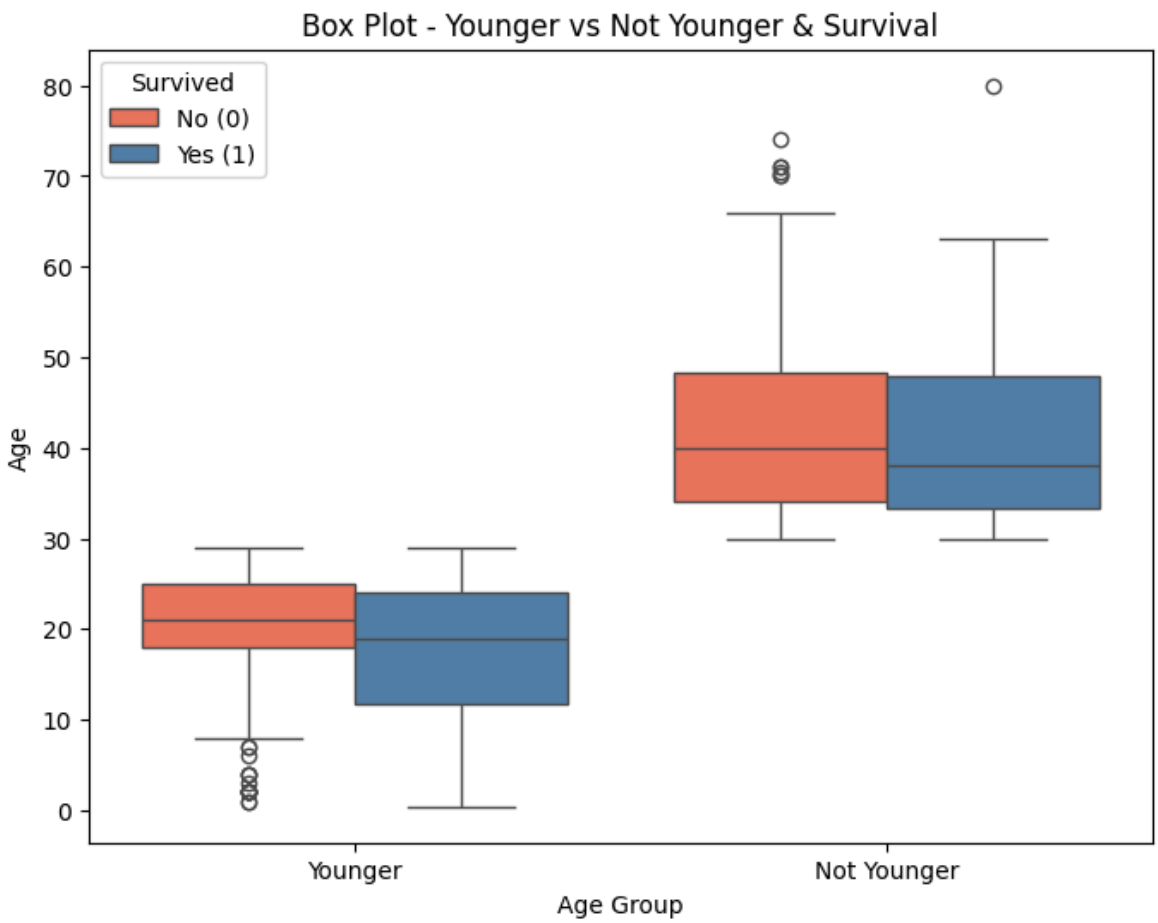
```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True

Shape: (891, 15)

Columns: ['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare', 'embarke
d', 'class', 'who', 'adult_male', 'deck', 'embark_town', 'alive', 'alone']
```

In [4]:
```
plt.figure(figsize=(8, 6))
ax = sns.boxplot(x='sex', y='age', data=dataset, hue='survived',
            palette={0: 'tomato', 1: 'steelblue'})

handles, labels = ax.get_legend_handles_labels()
ax.legend(handles=handles, labels=['No (0)', 'Yes (1)'], title='Survived')

plt.title("Box Plot - Age Distribution by Gender & Survival")
plt.xlabel("Gender")
plt.ylabel("Age")
plt.show()
```



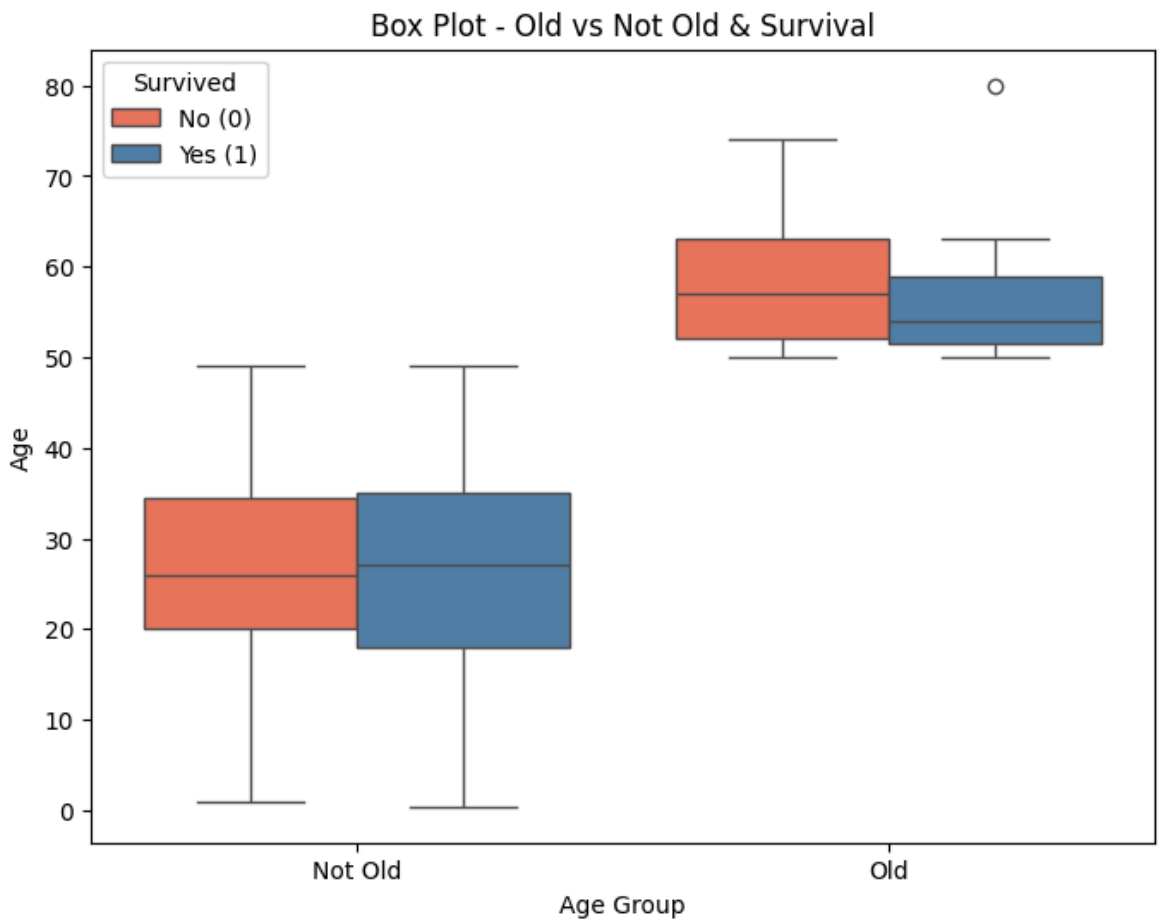
In [5]:
```
dataset['is_younger'] = dataset['age'].apply(
    lambda x: 'Younger' if x < 30 else 'Not Younger'
)
```

In [11]:
```
plt.figure(figsize=(8, 6))
ax = sns.boxplot(x='is_younger', y='age', data=dataset, hue='survived',
            palette={0: 'tomato', 1: 'steelblue'})

handles, labels = ax.get_legend_handles_labels()
ax.legend(handles=handles, labels=['No (0)', 'Yes (1)'], title='Survived')

plt.title("Box Plot - Younger vs Not Younger & Survival")
plt.xlabel("Age Group")
plt.ylabel("Age")
plt.show()
```

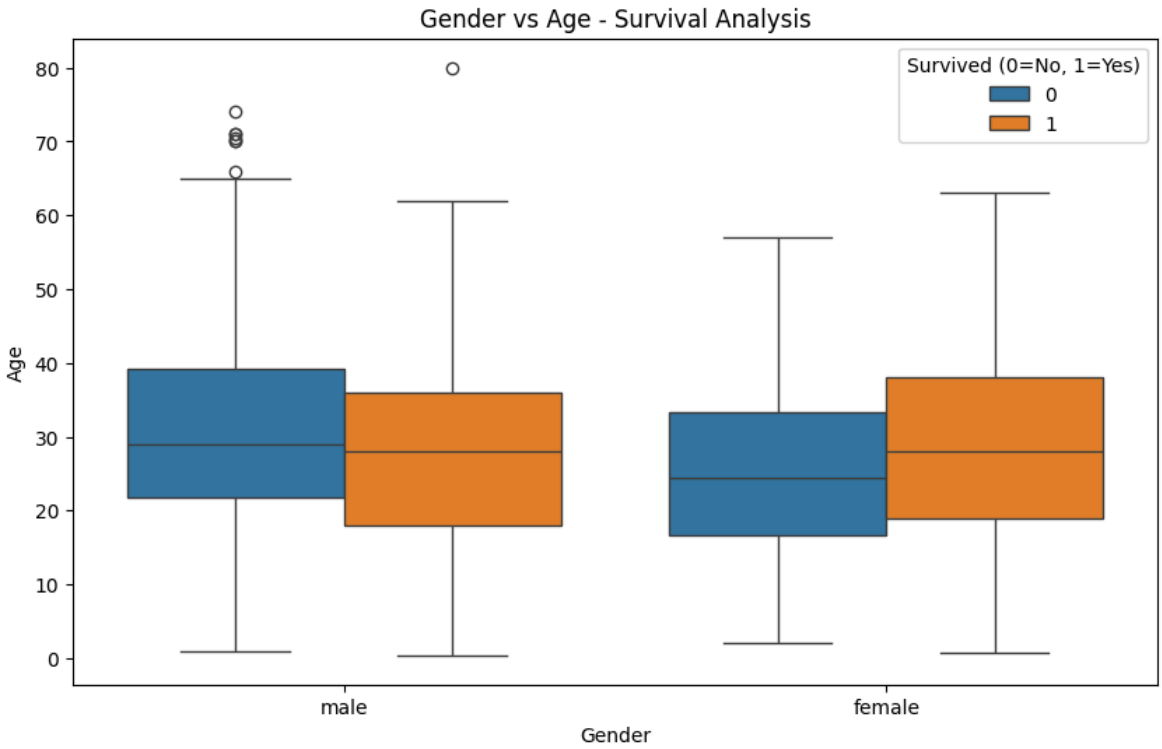


In [7]:
```
dataset['is_old'] = dataset['age'].apply(
    lambda x: 'Old' if x >= 50 else 'Not Old'
)
```

In [14]:
```
plt.figure(figsize=(8, 6))
ax = sns.boxplot(x='is_old', y='age', data=dataset, hue='survived',
            palette={0: 'tomato', 1: 'steelblue'})

handles, labels = ax.get_legend_handles_labels()
ax.legend(handles=handles, labels=['No (0)', 'Yes (1)'], title='Survived')

plt.title("Box Plot - Old vs Not Old & Survival")
plt.xlabel("Age Group")
plt.ylabel("Age")
plt.show()
```



In [9]:
```
plt.figure(figsize=(10, 6))
sns.boxplot(x='sex', y='age', data=dataset, hue='survived')
plt.title("Gender vs Age - Survival Analysis")
plt.xlabel("Gender")
plt.ylabel("Age")
plt.legend(title='Survived (0=No, 1=Yes)')
plt.show()
```



Observation:

From the analysis, we can see that gender had a bigger effect on survival than age. Females were more likely to survive compared to males in almost every age group. Age did have some impact, as younger passengers had slightly better chances of survival, while older people had lower chances. However, the difference based on age was not very strong. The graphs also show that both survivors and non-survivors had similar age ranges, so age alone was not a clear deciding factor. Overall, gender played the most important role, while age had only a small influence on survival.